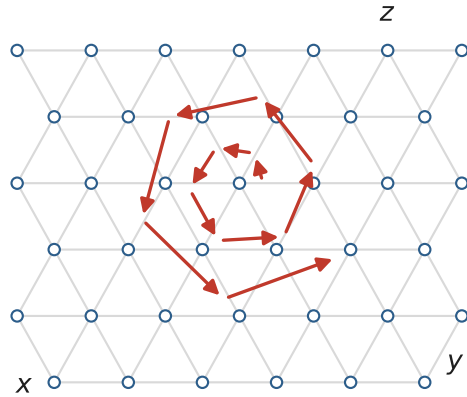


The Gauge Lift

1 6D Lattice Deformation



Phason displacement $u_{\perp}$ in internal space $E_{\perp}$.

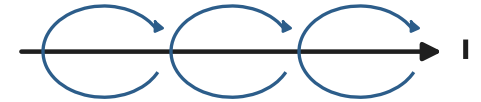
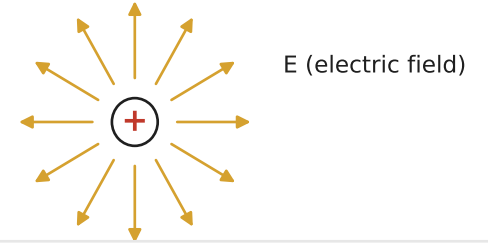
2 Lift Operation: Curl Extraction

$$F_{\mu\nu} \propto \nabla \times u_{\perp} \quad \longrightarrow$$

Internal space ($E_{\perp}$) $\rightarrow$ external gauge field.

Curl extraction selects the transverse vector content.

3 Electromagnetic Field $F_{\mu\nu}$ in 3D Spacetime



B (magnetic field)

$$B = \nabla \times u_{\perp}$$

$$E = \partial_t u_{\perp}$$

Spin-1: vector field structure inherited from curl operation. Massless: $U(1)$ gauge protection.